

Chapter 5

Derivatives

§ Section: 2.7-2.8

5.1 Derivatives

The problem of finding the tangent line to a curve and the problem of finding the velocity of an object both involve finding the same type of limit. This special type of limit is called a **derivative**.

Definition:

The **tangent line** to the curve $y = f(x)$ at the point $P(a, f(a))$ is the line through P with slope

$$m = \lim_{x \rightarrow a} \frac{f(x) - f(a)}{x - a}$$

provided that this limit exists.

Remark:

Alternatively, m can be defined as $m = \lim_{h \rightarrow 0} \frac{f(a+h) - f(a)}{h}$.

Definition:

The **derivative of a function f at a number a** , denoted by $f'(a)$, is

$$f'(a) = \lim_{h \rightarrow 0} \frac{f(a+h) - f(a)}{h}$$

if this limit exists.

Remark:

Similarly, $f'(a)$ can be defined as $f'(a) = \lim_{x \rightarrow a} \frac{f(x) - f(a)}{x - a}$.

Suppose y is a quantity that depends on another quantity x . Thus y is a function of x and we write $y = f(x)$. Then the derivative of f at $x = a$ means the **instantaneous rate of change** of y with respect to x at $x = a$.

Instantaneous rate of change

$$\lim_{\Delta x \rightarrow 0} \frac{\Delta y}{\Delta x} = \lim_{x \rightarrow a} \frac{f(x) - f(a)}{x - a}$$

Example 1.1

$$f'(1) = \frac{2}{3}$$

Find $f'(1)$ where $f(x) = \sqrt[3]{x^2}$. Find the tangent line of $y = \sqrt[3]{x^2}$ at $x = 1$.

$$\begin{aligned} \text{sol: } f'(1) &= \lim_{x \rightarrow 1} \frac{f(x) - f(1)}{x - 1} = \lim_{x \rightarrow 1} \frac{\sqrt[3]{x^2} - 1}{x - 1} \\ &= \lim_{x \rightarrow 1} \frac{(\sqrt[3]{x^2} - 1)(\sqrt[3]{x^4} + \sqrt[3]{x^2} + 1)}{(x - 1)(\sqrt[3]{x^4} + \sqrt[3]{x^2} + 1)} = \lim_{x \rightarrow 1} \frac{x^2 - 1}{(x - 1)(\sqrt[3]{x^4} + \sqrt[3]{x^2} + 1)} \\ &= \lim_{x \rightarrow 1} \frac{x + 1}{(\sqrt[3]{x^4} + \sqrt[3]{x^2} + 1)} = \frac{2}{3} \end{aligned}$$

The tangent line is
 $y - 1 = \frac{2}{3}(x - 1)$
 $\Rightarrow y = \frac{2}{3}x + \frac{1}{3}$

Example 1.2

Find $f'(0)$, where $f(x) = \sqrt[3]{x^2}$.

$$\begin{aligned} \text{sol: } f'(0) &= \lim_{x \rightarrow 0} \frac{f(x) - f(0)}{x - 0} = \lim_{x \rightarrow 0} \frac{\sqrt[3]{x^2} - 0}{x - 0} \\ &= \lim_{x \rightarrow 0} \frac{1}{\sqrt[3]{x}} \quad \text{DNE.} \end{aligned}$$

Example 1.3

Find $f'(0)$, where $f(x) = \begin{cases} x \sin\left(\frac{1}{x}\right), & \text{if } x \neq 0 \\ 0, & \text{if } x = 0 \end{cases}$.

$$\begin{aligned} \text{sol: } f'(0) &= \lim_{x \rightarrow 0} \frac{f(x) - f(0)}{x - 0} \\ &= \lim_{x \rightarrow 0} \frac{x \sin\left(\frac{1}{x}\right)}{x} = \lim_{x \rightarrow 0} \sin\left(\frac{1}{x}\right) \quad \text{DNE} \end{aligned}$$

Exercise 1.1

$$f(x) = \begin{cases} x^2 \cos\left(\frac{1}{x}\right), & \text{if } x \neq 0 \\ 0, & \text{if } x = 0 \end{cases}. \text{ Find } f'(0). = \lim_{x \rightarrow 0} \frac{f(x) - f(0)}{x - 0} = \lim_{x \rightarrow 0} x \cos\left(\frac{1}{x}\right) = 0$$

$$\lim_{h \rightarrow 0} \frac{a^h - a^0}{h} = f'(0), \text{ where } f(x) = a^x.$$

Example 1.4

Recognize $\lim_{x \rightarrow 1} \frac{\ln x}{x - 1}$, $\lim_{h \rightarrow 0} \frac{a^h - 1}{h}$, $\lim_{x \rightarrow 0} \frac{\sin x}{x}$ as derivatives.

$$\lim_{x \rightarrow 0} \frac{\sin x}{x} = \lim_{x \rightarrow 0} \frac{\sin x - \sin 0}{x - 0} = f'(0), \text{ where } f(x) = \sin x.$$

$$\lim_{x \rightarrow 1} \frac{\ln x}{x - 1} = \lim_{x \rightarrow 1} \frac{\ln x - \ln 1}{x - 1} = f'(1), \text{ where } f(x) = \ln x$$

5.2 Differentiable functions

Let x be a variable. We can define

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}.$$

Now we regard $f'(x)$ as a new function, called the **derivative** of f .

Notation

$$f'(x) = y' = \frac{dy}{dx} = \frac{df}{dx} = \frac{d}{dx} f(x) = Df(x) = D_x f(x)$$

The symbols D and $\frac{d}{dx}$ are called **differentiation operators** because they indicate the operation of *differentiation*.

If we want to indicate the value of a derivative in Leibniz notation at a specific number a , we use the notation $\left. \frac{dy}{dx} \right|_{x=a}$ or $\left. \frac{dy}{dx} \right]_{x=a}$ which is a synonym for $f'(a)$.

Example 2.1

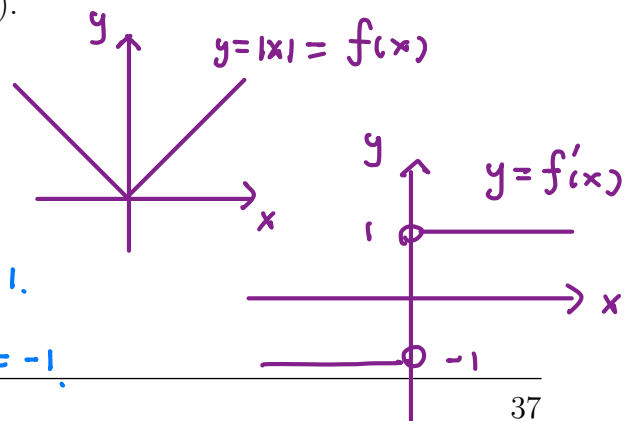
Find $f'(x)$ where $f(x) = |x|$.

$$\text{Sol: } f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} = \lim_{h \rightarrow 0} \frac{|x+h| - |x|}{h}$$

$$(i) \text{ If } x > 0, \text{ then } f'(x) = \lim_{h \rightarrow 0} \frac{(x+h) - x}{h} = \lim_{h \rightarrow 0} \frac{h}{h} = 1.$$

$$(ii) \text{ If } x < 0, \text{ then } f'(x) = \lim_{h \rightarrow 0} \frac{-(x+h) - (-x)}{h} = \lim_{h \rightarrow 0} \frac{-h}{h} = -1.$$

$$(iii) f'(0) = \lim_{h \rightarrow 0} \frac{|h|}{h} \text{ DNE. } f'(x) = \begin{cases} 1, & \text{if } x > 0 \\ -1, & \text{if } x < 0 \end{cases}$$



Definition:

A function f is **differentiable at a** if $f'(a)$ exists. It is **differentiable on an open interval (a, b)** [or (a, ∞) or $(-\infty, a)$ or $(-\infty, \infty)$] if it is differentiable at every number in the interval.

Theorem:

If f is differentiable at a , then f is continuous at a .

Remark:

The converse of Theorem is false; that is, there are functions that are continuous but not differentiable.

Example 2.2

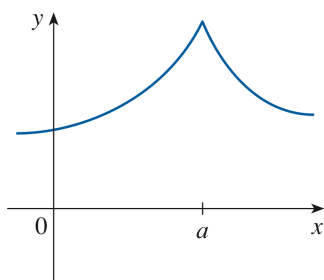
Show that if $f(x)$ is differentiable at $x = a$, then $f(x)$ is continuous at $x = a$.

Sol: $\because f$ is diff at $x=a \quad \therefore \lim_{x \rightarrow a} \frac{f(x) - f(a)}{x - a} = f'(a)$ exists.

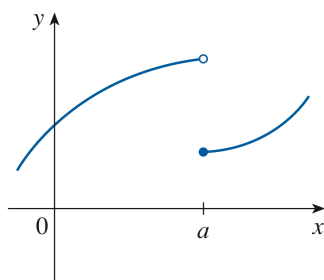
Hence $\lim_{x \rightarrow a} f(x) = \lim_{x \rightarrow a} \left(\frac{f(x) - f(a)}{x - a} \right) \cdot (x - a) + f(a) = f'(a) \cdot 0 + f(a) = f(a).$

Some cases of not differentiable functions

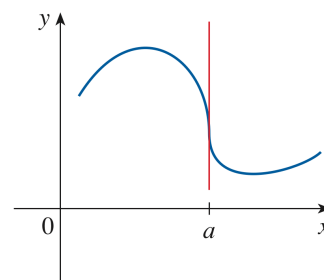
A discontinuous point, A corner, A vertical tangent, An oscillating point



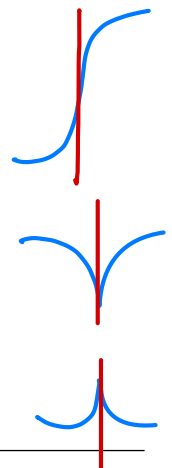
(a) A corner



(b) A discontinuity



(c) A vertical tangent



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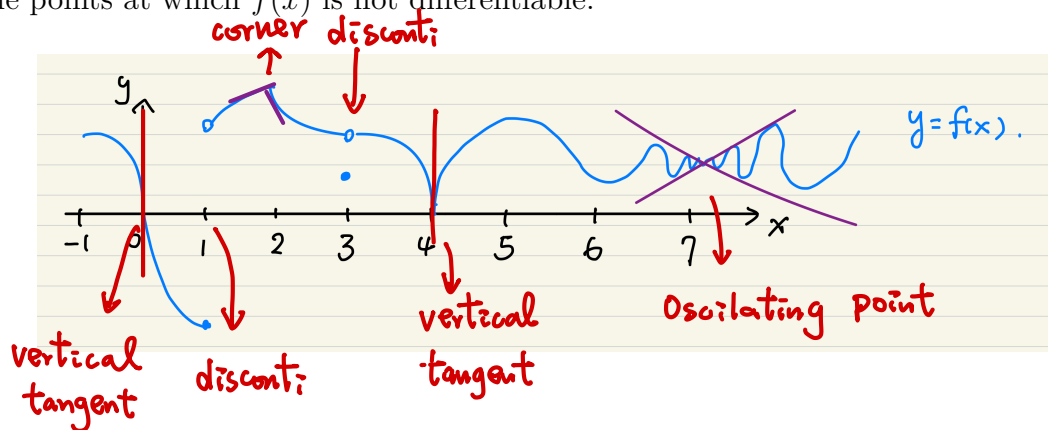
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Example 2.3

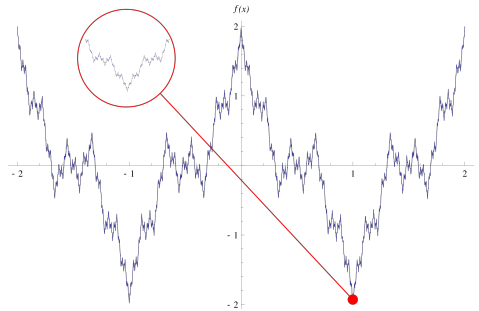
Specify the points at which $f(x)$ is not differentiable.



Continuous Everywhere but Differentiable Nowhere: Weierstrass Functions

$$f(x) = \sum_{n=0}^{\infty} a^n \cos(b^n \pi x)$$

where $0 < a < 1$, b is a positive odd integer and $ab > 1 + \frac{3}{2}\pi$.



5.3 Higher Derivatives

If $f(x)$ is a differentiable function, then its derivative f' is also a function, so f' may have a derivative of its own, denoted by $(f')' = f''$. This new function f'' is called the **second derivative** of f .

Notations for the Second Derivatives

$$f'' = \frac{d}{dx} \left(\frac{dy}{dx} \right) = \frac{d^2 y}{dx^2}$$

Higher derivatives of $f(x)$ can be defined in a similar way:

Notations for Higher Derivatives

$$y''' = f'''(x) = \frac{d}{dx} \left(\frac{d^2 y}{dx^2} \right) = \frac{d^3 y}{dx^3}$$

$$y^{(n)} = f^{(n)}(x) = \frac{d^n y}{dx^n}$$

5.4 Sketch Derivatives from the Graphs of Functions

The derivative of $f(x)$ gives the slope of the tangent line to the curve $y = f(x)$. Therefore, if $f(x)$ is increasing on an interval I , then $f'(x) > 0$ on I . Conversely, if $f(x)$ is decreasing on an interval I , then $f'(x) < 0$ on I . Furthermore, $f'(a) = 0$ whenever $y = f(x)$ has a horizontal tangent line at the point $(a, f(a))$.

Example 4.1

The graph of $f(x)$ is given as below. Sketch the graph of $f'(x)$.

